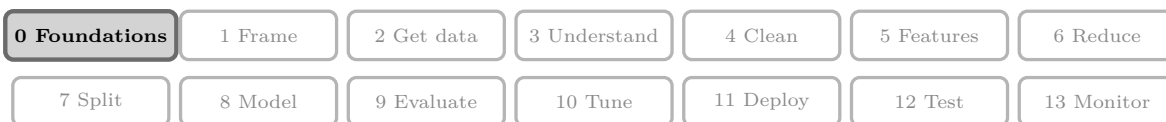


Matrix Multiplication as Composition

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Dot product ([Note MA-050](#)); Linear transformations and matrices ([Note MA-053](#)).
- **Leads to:** Jacobian and matrix gradients ([Note MA-063](#)); Neural networks ([Note DL-002](#)); Forward propagation ([Note DL-010](#)).

1. Overview

Key point: Multiplying two matrices gives the matrix of “apply one transformation, then the other”; we read the product from right to left.

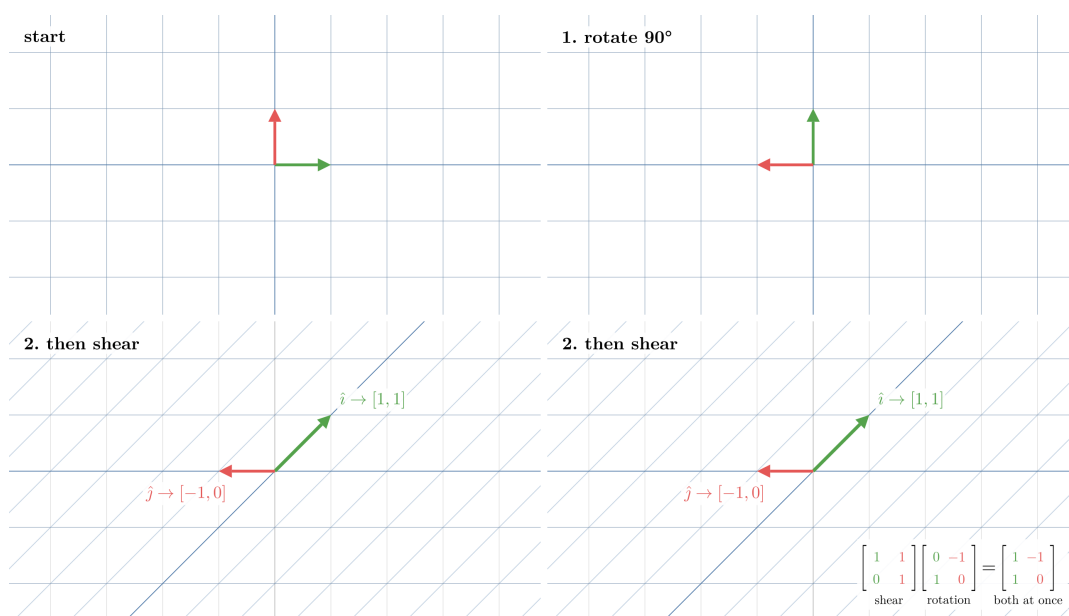


Figure 1: Rotating the plane by 90° and then shearing it is one single transformation; its matrix is the product of the two matrices

Figure 1 applies two transformations one after the other: a rotation by 90° , then a shear. From start to finish, the plane has undergone a third linear transformation, with a matrix of its own. That matrix is the product of the shear matrix and the rotation matrix.

This Note builds on the [linear transformations and matrices Note](#): the columns of a matrix are where $\hat{i}$ and $\hat{j}$ land, and $A\mathbf{x}$ is the linear combination of those columns.

2. Composition: one transformation after another

Key point: Applying one linear transformation and then another is again a linear transformation, called their composition; following $\hat{i}$ and $\hat{j}$ gives its matrix.

2.1 Turning a stage twice

Key point: Two turns of a stage, one after the other, are the same as one combined turn; the matrix of that combined turn is the product of the two turn matrices.

Picture a concert stage with x and y axes centred on it. A seat sits at $(2, 1)$. Now the stage turns, and every seat turns with it (Figure 2):

1. **First turn, 180° .** The seat moves to $(-2, -1)$.
2. **Second turn, 90° clockwise.** The seat moves to $(-1, 2)$.

Each turn is a linear transformation (see the [linear transformations and matrices Note](#)), so each has a matrix whose columns are where $\hat{i}$ and $\hat{j}$ land. The 180° turn sends $\hat{i}$ to $[-1, 0]$ and $\hat{j}$ to $[0, -1]$; the 90° clockwise turn sends $\hat{i}$ to $[0, -1]$ and $\hat{j}$ to $[1, 0]$:

$$T_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad T_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Done the long way, the seat needs two multiplications. Each entry of the result is two products added, one entry per line.

First turn, $T_1[2, 1]$:

$$\text{top} = (-1)(2) + (0)(1) = -2 + 0 = -2$$

$$\text{bottom} = (0)(2) + (-1)(1) = 0 - 1 = -1$$

Second turn, $T_2[-2, -1]$:

$$\text{top} = (0)(-2) + (1)(-1) = 0 - 1 = -1$$

$$\text{bottom} = (-1)(-2) + (0)(-1) = 2 + 0 = 2$$

So the seat goes $[2, 1] \rightarrow [-2, -1] \rightarrow [-1, 2]$. Done the short way, we first combine the two turns into one matrix, the product T_2T_1 . Each entry is a row of T_2 times a column of T_1 : the first entry of the row times the first entry of the column, plus the second times the second.

$$\text{row 1, column 1} = (0)(-1) + (1)(0) = 0 + 0 = 0$$

$$\text{row 1, column 2} = (0)(0) + (1)(-1) = 0 - 1 = -1$$

$$\text{row 2, column 1} = (-1)(-1) + (0)(0) = 1 + 0 = 1$$

$$\text{row 2, column 2} = (-1)(0) + (0)(-1) = 0 + 0 = 0$$

$$T_2 T_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Applied to the seat:

$$(0)(2) + (-1)(1) = -1, \quad (1)(2) + (0)(1) = 2$$

so $T_2 T_1[2, 1] = [-1, 2]$, the same spot in one step.

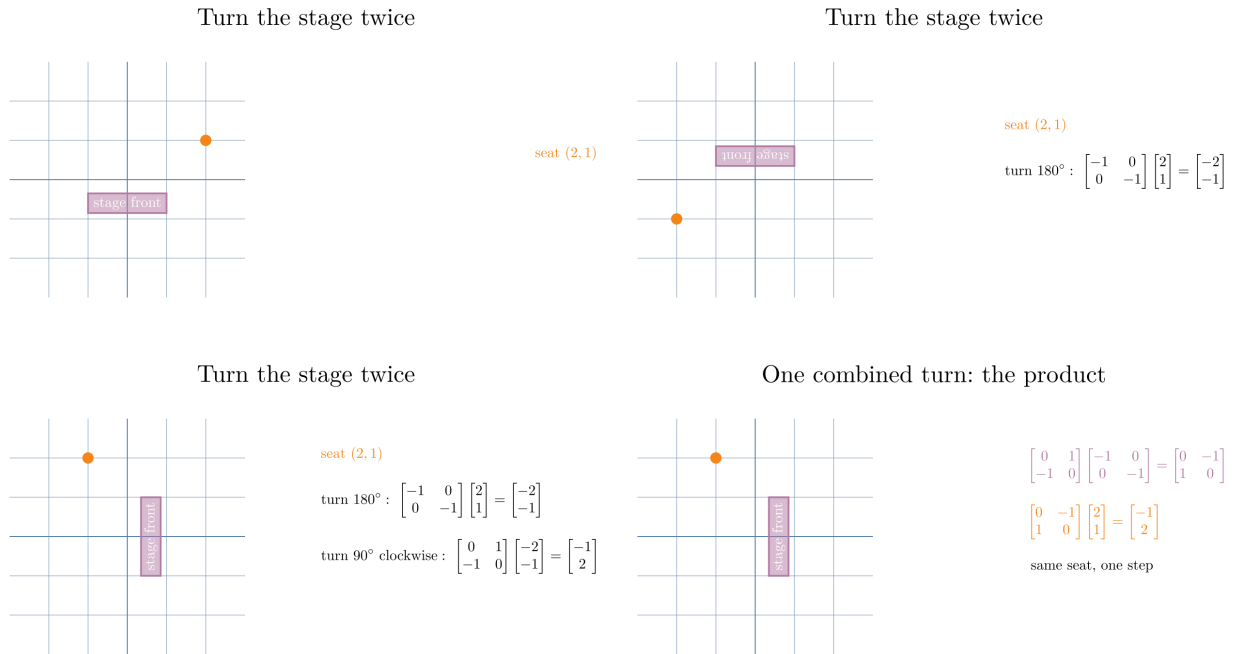


Figure 2: A seat on a turning stage: two turns move it from (2, 1) to (-2, -1) to (-1, 2); the product of the two turn matrices, a single 90° counterclockwise turn, sends it there in one step. Story after StatQuest, “Essential Matrix Algebra for Neural Networks”

In Figure 2, compare the last frame with the third: the combined turn lands the seat on the same spot, (-1, 2), in one step. However many times the stage turns, multiplying the turn

matrices gives one combined turn. That is the reason matrix multiplication is defined the way it is: the product is built so that it combines a sequence of transformations into one. Sections 2.2 to 4 show how to compute it.

2.2 Following the basis vectors

Key point: To find the matrix of a composition, follow $\hat{i}$ and $\hat{j}$ through both steps; where they finally land are its columns.

We often want to apply one transformation and then another. The overall effect is again a linear transformation (grid lines are still parallel and evenly spaced, and the origin has not moved). The overall effect is called the **composition** (G-431) of the two transformations.

Like any linear transformation, the composition has a matrix, found by following $\hat{i}$ and $\hat{j}$ to their final places. In Figure 1, a rotation is followed by a shear:

- **Rotation by 90°** has the matrix R with columns $[0, 1]$ and $[-1, 0]$: it sends $\hat{i}$ to $[0, 1]$ and $\hat{j}$ to $[-1, 0]$.
- **The shear** has the matrix S with columns $[1, 0]$ and $[1, 1]$: it keeps $[1, 0]$ fixed and sends $[0, 1]$ to $[1, 1]$.
- **After both**, $\hat{i}$ has ended at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$.

So the composition has the matrix with those two columns:

$$SR = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Each entry is a row of S times a column of R , one per line:

$$\text{row 1, column 1} = (1)(0) + (1)(1) = 0 + 1 = 1$$

$$\text{row 1, column 2} = (1)(-1) + (1)(0) = -1 + 0 = -1$$

$$\text{row 2, column 1} = (0)(0) + (1)(1) = 0 + 1 = 1$$

$$\text{row 2, column 2} = (0)(-1) + (1)(0) = 0 + 0 = 0$$

$$SR = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

The columns $[1, 1]$ and $[-1, 0]$ are where $\hat{i}$ and $\hat{j}$ ended.

This one matrix does in a single step what the rotation and the shear do in two.

3. Why the product is written right to left

Key point: In $SR\mathbf{x}$, the matrix next to the vector acts first: R is applied first, then S .

Applying the rotation and then the shear to a vector $\mathbf{x}$ the long way means two multiplications: first $R\mathbf{x}$, then S times that result, $S(R\mathbf{x})$. The composition matrix must give the same answer for every $\mathbf{x}$, since it describes the same overall movement. So it is natural to call it the **product** (G-1179) SR :

$$S(R\mathbf{x}) = (SR)\mathbf{x}$$

Note the order. The transformation on the right (R) happens first and the one on the left (S) second.

Read a product from right to left

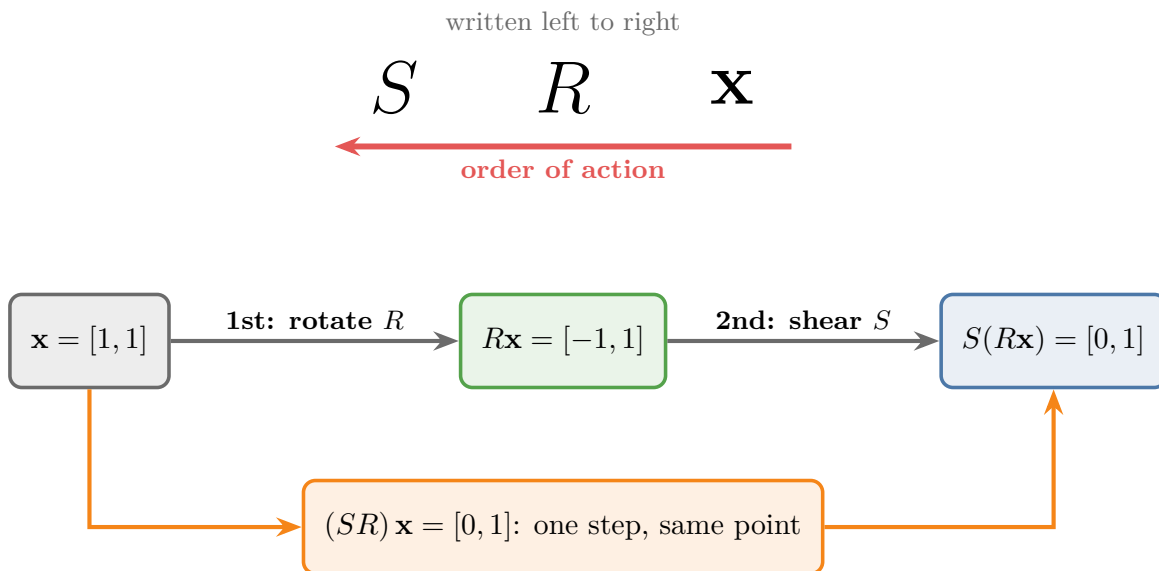


Figure 3: The product is written left to right but acts right to left; the vector $[1, 1]$ goes through R , then S , and SR takes it to the same point in one step

In Figure 3, follow $[1, 1]$: R turns it into $[-1, 1]$, S then gives $[0, 1]$, and SR gives $[0, 1]$ in one step. The right-to-left order comes from function notation: in $f(g(x))$, g acts first. A matrix product is read from right to left.

4. Computing a product column by column

Key point: The first column of M_2M_1 is M_2 times the first column of M_1 ; the second column is M_2 times the second column of M_1 .

We can find the product without watching any animation, from the numbers alone. Take

$$M_1 = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

and apply M_1 first, then M_2 .

1. **In words:** after M_1 , $\hat{i}$ sits at the first column of M_1 . Applying M_2 to that column gives where $\hat{i}$ finally lands: the first column of the product. Do the same with the second column for $\hat{j}$.
2. **Formula:**

$$M_2M_1 = \left[M_2 \cdot (\text{column 1 of } M_1) \mid M_2 \cdot (\text{column 2 of } M_1) \right]$$

3. **Example:**

- $\hat{i}$ first lands on $[1, 1]$. Then $M_2[1, 1] = 1 \cdot [0, 1] + 1 \cdot [2, 0] = [2, 1]$.
- $\hat{j}$ first lands on $[-2, 0]$. Then $M_2[-2, 0] = -2 \cdot [0, 1] + 0 \cdot [2, 0] = [0, -2]$.
- Check one entry, row 1 of M_2 times column 1 of M_1 : $(0)(1) + (2)(1) = 0 + 2 = 2$, the top-left entry below.

$$M_2M_1 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

Figure 4 runs this example on the grid. Watch $\hat{i}$ (green) and $\hat{j}$ (red): after M_1 they sit on the columns of M_1 , and after M_2 each one's final position fills one column of the product. Following the basis vectors like this is the approach of Sanderson's *Essence of Linear Algebra*, chapter 4 (3Blue1Brown).

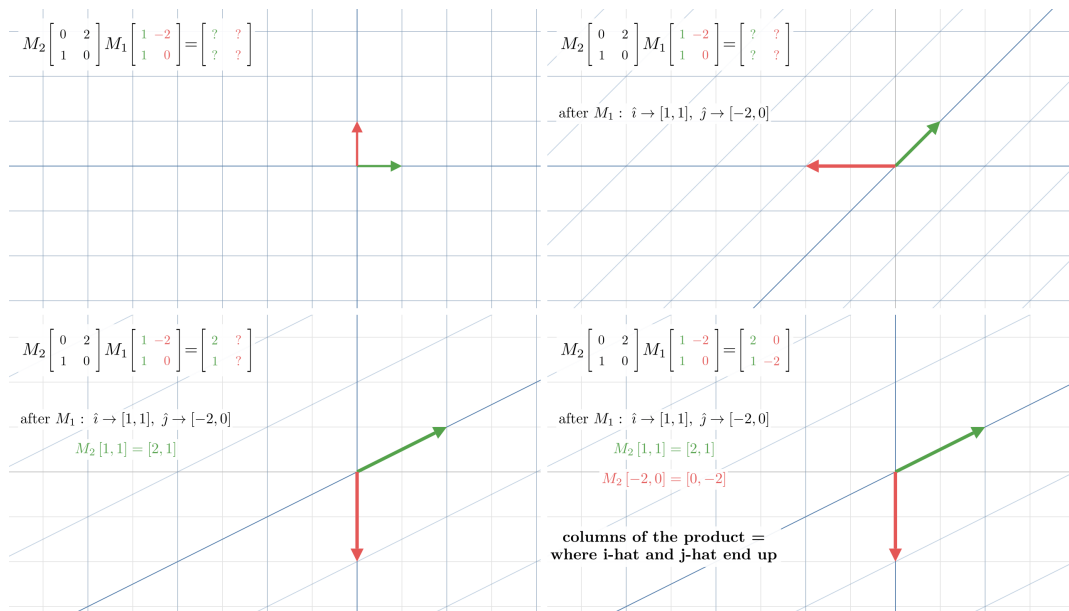


Figure 4: The product M_2M_1 built column by column: M_1 moves the basis vectors to its columns, M_2 then moves them to $[2, 1]$ and $[0, -2]$, the columns of the product

The same reasoning with letters gives the general rule. Take

$$M_1 = \begin{bmatrix} e & f \\ g & h \end{bmatrix}, \quad M_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The first column of the product is $e[a, c] + g[b, d]$ and the second is $f[a, c] + h[b, d]$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Extra: The same formula read entry by entry is the school rule “row times column”. The entry in row i , column j of AB is the dot product of row i of A with column j of B . In the example, row 1 of M_2 is $[0, 2]$ and column 1 of M_1 is $[1, 1]$, and $0 \times 1 + 2 \times 1 = 2$, the top-left entry. The row-times-column view is why the [dot product and cosine similarity Note](#) calls the dot product the building block of matrix multiplication.

Python: @ multiplies two matrices.

```
import numpy as np
```

```
M1 = np.array([[1, -2], [1, 0]])
M2 = np.array([[0, 2], [1, 0]])
M2 @ M1          # [[2, 0], [1, -2]]
x = np.array([1, 1])
M2 @ (M1 @ x)    # [2, -1]: two steps
```


5. Order matters

Key point: In general $AB \neq BA$: shearing then rotating is a different transformation from rotating then shearing.

Does the order of the two matrices matter? Thinking in transformations, we can answer by picturing them. Figure 5 runs the rotation R and the shear S of Section 2.2 in both orders at once; watch the two pairs of arrows end up in different places.

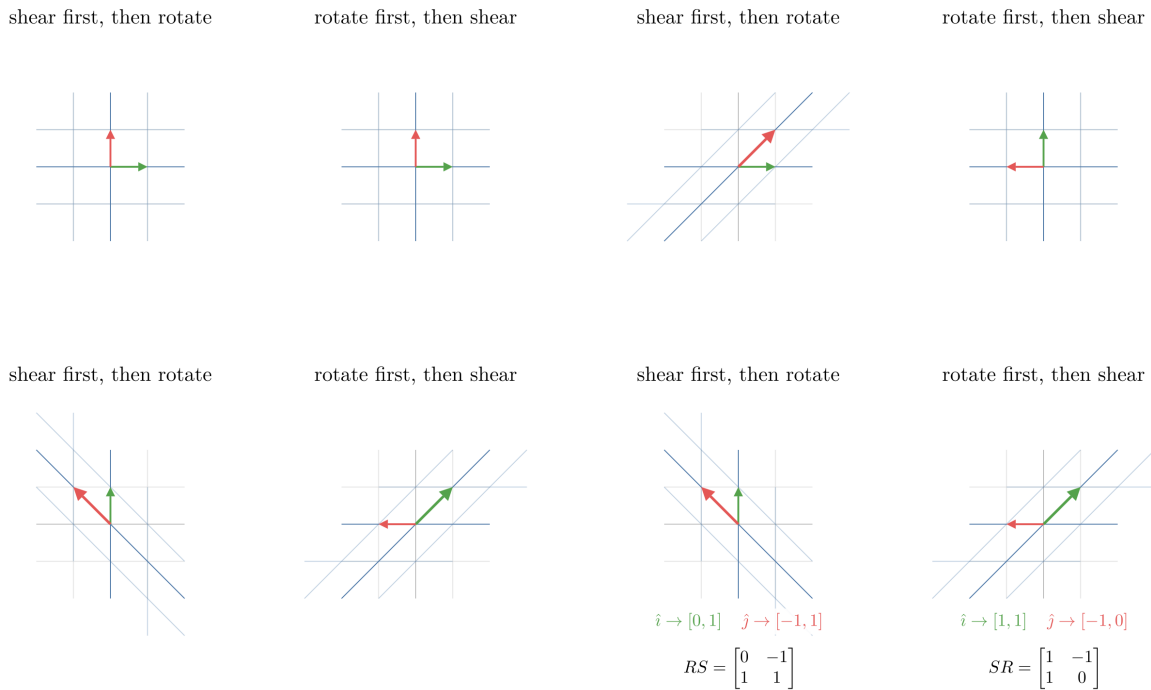


Figure 5: The same two transformations in the two possible orders, side by side; at the end, compare where $\hat{i}$ (green) and $\hat{j}$ (red) have landed. Example after 3Blue1Brown, “Matrix multiplication as composition”

- **Shear first, then rotate** (RS): $\hat{i}$ ends at $[0, 1]$ and $\hat{j}$ at $[-1, 1]$. The two vectors point close together.
- **Rotate first, then shear** (SR): $\hat{i}$ ends at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$. They point far apart.

The matrix SR was computed in section 2.2. The matrix RS , entry by entry (row of R times column of S):

$$\text{row 1, column 1} = (0)(1) + (-1)(0) = 0$$

$$\text{row 1, column 2} = (0)(1) + (-1)(1) = -1$$

$$\text{row 2, column 1} = (1)(1) + (0)(0) = 1$$

$$\text{row 2, column 2} = (1)(1) + (0)(1) = 1$$

The overall effects differ, so $RS \neq SR$:

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = SR$$

Matrix multiplication is **not commutative** (G-1352), unlike the multiplication of numbers. In code this means $A @ B$ and $B @ A$ are, in general, different matrices.

6. Grouping does not matter

Key point: $(AB)C = A(BC)$, because both mean “apply C , then B , then A ”.

For three matrices A , B and C , we can first compute AB and multiply the result by C , or first compute BC and put A in front. Both give the same matrix:

$$(AB)C = A(BC)$$

This property is called **associativity** (G-219). Proving it by expanding all the entries is long and tells us nothing. Seen as transformations it is immediate: both sides mean “apply C , then B , then A ”, the same three steps in the same order. Only the bookkeeping differs.

Python: A quick check with the three matrices of this Note.

```
R = np.array([[0, -1], [1, 0]])
S = np.array([[1, 1], [0, 1]])
np.array_equal((S @ R) @ M1, S @ (R @ M1))    # True
np.array_equal(S @ R, R @ S)                  # False
```

7. Where ML multiplies matrices

Key point: A product needs matching inner sizes; stacked linear layers collapse into one matrix, which is why neural networks need activation functions between them.

7.1 The shape rule

Key point: An $m \times n$ matrix times an $n \times p$ matrix gives an $m \times p$ matrix; the inner sizes must match.

So far all matrices were 2×2 . In ML the matrices are rectangular, and the column-by-column view explains the shape rule. In AB , each column of B is a vector that A transforms, so each column of B must have as many entries as A has columns.

1. **In words:** the number of columns of the left matrix must equal the number of rows of the right matrix; the result takes the rows of the left and the columns of the right.

2. **Formula:**

$$(m \times n)(n \times p) = (m \times p)$$

3. **Example:** the data matrix X with 3 points and 2 **features** (G-772) (input variables, one column each) times A^T in the [linear transformations and matrices Note](#) (section 7.1) is $(3 \times 2)(2 \times 2) = 3 \times 2$: three transformed points. PCA's XW^T with 40 points, 3 features and 2 components is $(40 \times 3)(3 \times 2) = 40 \times 2$ (see the [PCA step by step Note](#), section 5.1).

A product with mismatched inner sizes does not exist. The turn matrix T_1 of Section 2.1 times the seat written as a row, $(2 \times 2)(1 \times 2)$, fails: each row of T_1 has two numbers, but each column of $[2 \ 1]$ has only one to pair them with. The dot product $a^T b$, $(1 \times n)(n \times 1) = 1 \times 1$, is the smallest case of the rule.

7.2 Points as rows: the same product, transposed

Key point: Libraries that store points as rows write $\mathbf{x}^T W^T$ instead of $W\mathbf{x}$; then the matrix applied first stands on the left.

This Note writes a point as a column and puts the matrix on its left, $W\mathbf{x}$. A data table stores each point as a row, so many libraries write the same step with the point on the left. Turning rows into columns is the **transpose** (G-2012); it flips a matrix across its diagonal, so the rows of W become the columns of W^T .

1. **In words:** in **row form** (G-2245), each point is a 1×2 row, and it is multiplied on the right by the transposed matrix.
2. **Example:** the stage of Section 2.1, with the seat as the row $[2 \ 1]$. $T_1^T = T_1$, so the first turn gives $[2 \ 1] T_1^T = [-2 \ -1]$. The second uses T_2^T , whose columns are $[0, 1]$ and $[-1, 0]$: row times first column is $-2 \times 0 + (-1) \times 1 = -1$, row times second column is $-2 \times (-1) + (-1) \times 0 = 2$. The seat lands on $[-1 \ 2]$, as before.
3. **Formula:** the two forms are transposes of each other,

$$(T_2 T_1 \mathbf{x})^T = \mathbf{x}^T T_1^T T_2^T$$

In row form the matrix applied first, T_1^T , stands next to the point on the left, and the product reads left to right.

PyTorch's linear layer uses the row form: `torch.nn.Linear` computes $y = xA^T + b$, with the input x as a row and the weights A stored one neuron per row (PyTorch docs, `torch.nn.Linear`). Both forms give the same numbers; only the bookkeeping differs.

Python: the two forms agree.

```
T1 = np.array([[ -1,  0], [ 0, -1]])
T2 = np.array([[ 0,  1], [-1,  0]])
T2 @ T1 @ np.array([ 2,  1])      # [-1, 2]: columns
np.array([[ 2,  1]]) @ T1.T @ T2.T # [[-1, 2]]: rows
```

7.3 Linear layers collapse into one

Key point: Two layers without an activation, $W_2(W_1\mathbf{x})$, are the single layer $(W_2W_1)\mathbf{x}$: stacking them adds no power.

A neural network applies layer after layer. Suppose two layers had no activation function between them, only the matrices W_1 and then W_2 . By Section 3, the two steps are one transformation:

$$W_2(W_1\mathbf{x}) = (W_2W_1)\mathbf{x}$$

With the numbers of Section 4, $W_1 = M_1$ and $W_2 = M_2$: the input $[1, 1]$ goes to $M_1[1, 1] = [-1, 1]$ and then to $M_2[-1, 1] = [2, -1]$. The single matrix M_2M_1 sends $[1, 1]$ straight to $[2, -1]$. However many linear layers we stack, the result is still one matrix, one linear transformation.

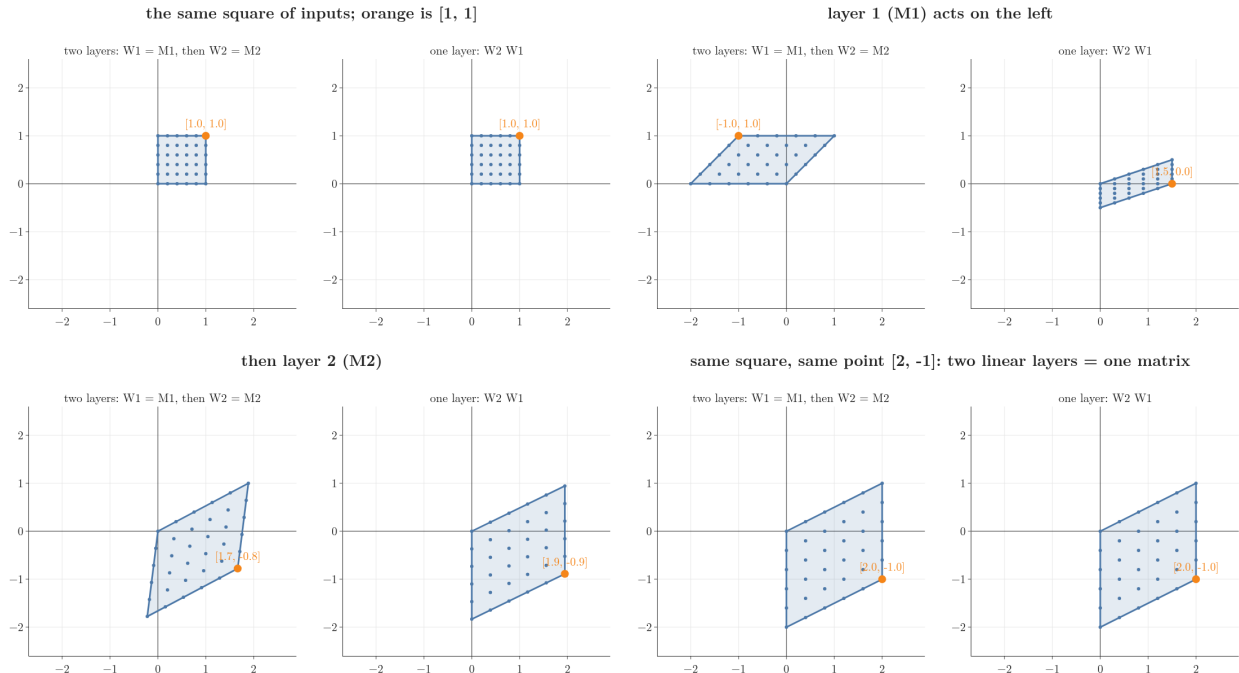


Figure 6: Two linear layers (left) and their single product matrix (right) move a square of inputs to the same place

In Figure 6, compare the two panels at the end: the same parallelogram, and $[1, 1]$ at $[2, -1]$ in both.

The collapse is why the hidden layers of a network end with a non-linear activation such as the [sigmoid](#). The activation bends the space between layers, so the composition can no longer be squeezed into a single matrix. The classic example is XOR: no linear model can fit it, while a network with one hidden layer and a non-linear activation can (Goodfellow et al. §6.1).

Extra: The same collapse shows in linear regression with engineered features. If every new feature is a linear combination of the old ones, the model can learn nothing new: the combined effect is still a linear combination of the original features. Polynomial features (see the [polynomial regression Note](#)) help precisely because squaring is not linear.

8. Summary

Idea	What it says	Example
Composition	one transformation after another	rotate, then shear
Product BA	matrix of “apply A , then B ”	$SR = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$
Column by column	column j of BA is B times column j of A	$M_2[1, 1] = [2, 1]$
Not commutative	$AB \neq BA$ in general	$RS \neq SR$
Associative	$(AB)C = A(BC)$	same three steps, same order
Transpose	turning the rows of a matrix into its columns	$(AB)^T = B^T A^T$
Row form	points as rows, multiplied by W^T on the right	$\mathbf{x}^T W^T$: the matrix applied first sits on the left
Shape rule	$(m \times n)(n \times p) = m \times p$	$(40 \times 3)(3 \times 2) = 40 \times 2$

- A matrix product is a composition of transformations, read from right to left.
- Follow $\hat{i}$ and $\hat{j}$ through both steps to get the columns of the product.
- Order matters; grouping does not.
- Stacked linear layers collapse into one matrix; activation functions prevent that.

9. Sources

Built from

- Starmer, J. (StatQuest), “Essential Matrix Algebra for Neural Networks, Clearly Explained!!!”, YouTube, <https://www.youtube.com/watch?v=ZTt9gsGcdDo>
- Sanderson, G. (3Blue1Brown), “Matrix multiplication as composition | Chapter 4, Essence of linear algebra”, 2016, [3blue1brown.com/lessons/matrix-multiplication](https://www.youtube.com/watch?v=XkY2DOUCWMU), <https://www.youtube.com/watch?v=XkY2DOUCWMU>

Other references

- PyTorch documentation, `torch.nn.Linear`, pytorch.org/docs/stable/generated/torch.nn.Linear.html.
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Section 6.1, learning XOR.

10. Key terms

Term	Meaning
Composition	Applying one transformation and then another, seen as one overall transformation

Term	Meaning
Matrix product	The matrix BA of the composition “apply A , then B ”
Not commutative	The order of the factors matters: $AB \neq BA$ in general
Associativity	$(AB)C = A(BC)$: the grouping of a product does not matter
Transpose	Turning the rows of a matrix into its columns; $(AB)^\top = B^\top A^\top$
Row form (G-2245)	Writing points as rows and multiplying $\mathbf{x}^\top W^\top$; the matrix applied first is on the left
Shape rule	$(m \times n)(n \times p) = m \times p$; the inner sizes must match